

Rakesh Thota

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PROFESSIONAL SUMMARY

Software Engineer with 4 years of experience designing and optimizing ETL pipelines, microservices, and cloud-native applications across AWS, GCP, and Azure. Implemented data solutions using Python, SQL, and modern data frameworks and BI tools (Power BI, Tableau, Looker). Proven track record of reducing processing latency, optimizing queries, and ensuring SLA compliance through automation, testing, and resilient software design that improved scalability, reliability, and performance of high-volume ETL workflows that empower organizations to make data-driven decisions with confidence

EDUCATION

University at Buffalo – State University of New York
Master of Science in Computer Science (AI/ML)

Buffalo, New York
Jan 2024 – May 2025

EXPERIENCE

SpringML Inc, Hyderabad, India

Jul 2022 – Dec 2023

Data Engineer

- Engineered and scaled end-to-end streaming **ETL pipelines** using **Google Cloud Pub/Sub & Dataflow**, processing 500k+ records daily with 40% lower latency, enabling near real-time analytics to business stakeholders.
- Orchestrated ML workflows with **Apache Airflow (Cloud Composer)** and **Vertex AI**, deploying inference results into **BigQuery**, boosting pipeline efficiency by ~25% and ensuring end-to-end reliability.
- Optimized **BigQuery** performance through partitioning, clustering, and schema optimization, cutting query execution time by 60% and compute costs by 40% across analytical workloads.
- Deployed CI/CD pipelines using **Cloud Build, Git, Docker**, and **GKE (Kubernetes)**, provisioned & managed infrastructure with **Terraform**, automating 55+ GCP resources, reducing manual effort by 80%, achieving more stable releases.
- Developed and deployed **REST APIs (Flask)** for secure client integrations, ensuring reliable delivery with OAuth2 and Secret Manager.
- Led proof-of-concept (POC) initiatives for financial document extraction automations using **Google Document AI, Camelot, OCR**, and complex **Regex** patterns, extracting key structured fields and eliminating manual data entry.

Persistent Systems, Hyderabad, India

Jan 2021 – Jun 2022

Associate Data Engineer

- Built and maintained scalable **AWS-based ETL pipelines** using **AWS Glue, Lambda, and Redshift**, processing multi-terabyte datasets daily and reducing transformation latency by 45%.
- Designed optimized data models and queries in **Redshift and SQL Server**, and integrated **SNS/SQS** with **RabbitMQ** for asynchronous job orchestration, cutting compute costs by 20%, speeding dashboards by 30%, and improving reliability.
- Collaborated with business and QA teams to define **unit and integration tests using pytest** for ETL pipelines, minimizing production incidents and improving system reliability.
- Partnered with client-facing teams to troubleshoot production incidents and implemented monitoring with **AWS CloudWatch**, improving issue resolution time by 35%.

Novartis, Hyderabad, India

Nov 2019 – Dec 2020

Junior Software Engineer

- Developed scalable data pipelines with **AWS Glue, S3, and Redshift**, processing 10TB+ clinical trial data monthly and reduced reporting latency by 20% and improving scalability.
- Built automated orchestration with Airflow on AWS (MWAA) and deployed REST APIs via API Gateway + Lambda to integrate clinical trial data pipelines with downstream applications.
- Generated **Tableau dashboards** to provide researchers with real-time experiment insights, reducing manual reporting by 40% and enabling faster data-driven insights.
- Collaborated with cross-functional teams (scientists, analysts, and engineers) to align pipelines with compliance requirements (under **HIPAA** and corporate governance rules), improving delivery timelines by 20%.

TECHNICAL SKILLS

- Programming Languages & APIs:** Python, SQL, Java, REST APIs
- Big Data Technologies & Cloud:** Apache Spark, Kafka, Airflow, Snowflake, Hadoop, Databricks, BigQuery, ETL/ELT, RabbitMQ
- Cloud & DevOps:** GCP, AWS (EC2, Lambda, CloudWatch, SQS, SNS), Azure (ADF, Synapse), Docker, Git, Terraform, Git
- Databases:** PostgreSQL, MySQL, Firebase, DynamoDB, SQL Server, MongoDB
- Analytics & Visualization:** Pandas, Tableau, Power BI, Looker, Excel
- AI/ML:** Vertex AI, Transformers, Tensorflow, PyTorch, NLP, Prompt Engineering, RAG pipelines
- Other:** Predictive Modeling, Data Warehousing, Data Modeling, Data Governance & Compliance (GDPR, HIPAA)

CERTIFICATIONS

- 5 x Google Cloud Certified: Professional Cloud Architect, Data Engineer, Cloud Developer, Cloud Security Engineer, and Associate Cloud Engineer, AWS Certified: Cloud Practitioner